

UL' YA, Russian Federation

WMO: 310870

Lat: 58.833N

Lon: 141.850E

Elev: 16

StdP: 14.69

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-17.8	-35.3	0.7	-19.8	-31.6	0.9	-16.5	41.1	-7.4	39.7	-3.6	20.1	320	0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	68.7	56.9	64.8	56.3	62.0	55.7	59.9	64.6	58.3	62.1	57.1	60.4	10.1	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.1	72.1	61.2	56.6	68.4	59.5	55.5	65.6	58.5	26.4	65.0	25.3	62.3	24.5	60.5	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.6	32.0	27.8	DB	-24.2	79.1	4.8	7.7	-27.7	84.6	-30.5	89.1	-33.2	93.4	-36.7	98.9	(4)
(5)			WB	-24.6	64.3	4.8	4.1	-28.0	67.3	-30.8	69.7	-33.5	72.1	-37.0	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.9	-4.2	0.8	11.5	24.5	35.7	46.0	53.7	55.4	46.8	30.3	9.1	-0.2	
	DBStd	22.40	9.18	8.91	8.70	7.17	5.53	5.35	4.47	4.63	6.01	8.43	10.54	10.09	
	HDD50	9152	1680	1378	1194	765	446	146	14	9	129	610	1226	1555	
	HDD65	14265	2145	1798	1659	1215	910	571	351	300	545	1075	1676	2020	
	CDD50	366	0	0	0	0	1	25	129	177	34	0	0	0	
	CDD65	4	0	0	0	0	0	0	1	3	0	0	0	0	
	CDH74	45	0	0	0	0	0	4	24	17	0	0	0	0	
	CDH80	12	0	0	0	0	0	0	11	1	0	0	0	0	
Wind	WSAvg	10.8	17.2	14.0	9.0	7.7	7.5	6.8	6.5	6.6	8.4	11.4	16.9	18.1	
Precipitation	PrecAvg	25.6	0.9	0.5	0.9	1.5	1.9	2.4	3.5	3.5	3.7	3.3	2.3	1.1	
	PrecMax	34.1	2.6	1.5	3.7	3.3	5.6	4.3	6.4	8.7	7.6	6.9	8.3	4.9	
	PrecMin	14.9	0.1	0.1	0.1	0.1	0.3	0.6	1.1	1.3	1.2	0.7	0.4	0.1	
	PrecStd	4.7	0.7	0.3	0.8	0.8	1.0	0.9	1.4	1.7	2.0	1.5	1.8	1.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	26.9	33.5	44.7	60.4	70.7	73.7	76.1	66.4	52.0	37.6	28.1	
		MCWB	25.3	26.1	27.3	35.1	46.2	54.5	60.3	59.3	54.2	41.4	36.7	27.0	
	2%	DB	19.5	22.7	30.2	39.6	53.5	62.3	67.1	69.9	61.2	47.2	32.2	25.5	
		MCWB	17.4	21.0	25.1	32.2	43.7	53.2	57.8	57.1	53.5	40.2	31.1	25.0	
	5%	DB	13.8	17.8	28.0	36.6	48.7	58.0	63.2	65.8	58.5	44.3	29.7	20.6	
		MCWB	12.1	15.9	24.3	31.2	41.1	51.1	56.5	56.6	53.1	39.1	28.7	19.5	
	10%	DB	8.9	13.4	25.1	33.8	44.3	54.3	60.5	62.9	56.4	41.5	25.5	13.6	
		MCWB	7.1	10.7	21.8	29.7	38.9	49.8	56.1	56.3	52.1	37.5	24.2	11.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.6	26.1	28.9	35.9	47.9	57.2	62.6	62.6	58.4	45.2	36.9	27.6
			MCDB	26.3	26.9	31.2	41.4	56.3	65.0	70.0	69.5	62.3	47.5	37.7	27.9
2%		WB	17.5	21.4	26.7	33.9	44.6	54.3	59.9	60.3	56.0	42.2	31.7	24.9	
		MCDB	18.8	22.3	29.5	38.2	51.8	60.6	64.8	64.8	58.6	45.2	32.1	25.5	
5%		WB	12.1	16.0	24.2	31.9	41.9	52.1	58.0	59.0	54.6	40.1	28.7	19.3	
		MCDB	13.7	17.5	27.3	34.8	47.2	56.4	61.3	62.7	57.0	43.2	29.5	20.2	
10%		WB	7.0	10.8	22.0	30.6	39.7	50.3	56.6	57.7	53.1	38.2	24.4	11.6	
		MCDB	8.7	13.1	25.0	32.9	43.4	53.3	59.4	61.2	55.6	41.0	25.7	13.2	
Mean Daily Temperature Range		5% DB	MDBR	7.5	10.0	15.1	13.2	9.9	10.5	10.0	12.2	13.4	10.6	8.2	7.4
			MCDBR	11.2	11.7	15.9	15.2	20.0	18.7	16.9	18.9	16.0	14.3	8.3	9.9
	MCWBR		11.0	11.0	12.8	11.0	12.2	11.3	10.8	11.4	11.3	10.9	8.6	10.3	
	5% WB	MCDBR	11.3	11.4	15.1	12.5	17.2	15.9	13.8	14.4	13.1	12.1	8.4	10.3	
		MCWBR	11.2	11.0	12.9	9.9	11.4	10.5	9.6	10.5	10.3	10.2	8.9	10.5	
Clear-Sky Solar Irradiance	taub	0.232	0.268	0.326	0.431	0.460	0.435	0.488	0.428	0.350	0.322	0.264	0.221		
	taud	2.099	2.131	1.998	1.776	1.923	2.203	2.075	2.261	2.446	2.283	2.140	2.039		
	Ebn at Noon	188	238	251	238	243	254	235	243	249	214	169	147		
	Edh at Noon	15	25	39	58	54	42	46	36	25	22	15	11		
All-Sky Solar Radiation	RadAvg	192	534	1091	1589	1578	1637	1436	1215	884	521	232	105		
	RadStd	13	25	53	121	184	158	231	152	114	60	28	9		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page